

5	(a)	(i)	wire cuts magnetic field	1
			e.m.f induced when there is cutting of flux	1
	(ii)		induced e.m.f. may produce an effect which 'opposes' change causing it	1
			as direction of movement changes, so does induced e.m.f.	1
	(b)	(i)	$x_0 = 1.5 \text{ mV}$	1
			$\omega = 2\pi / T = 2\pi / (3 \times 10^{-3})$	1 (for formula/method)
			$= 2090 \text{ rad s}^{-1}$	1
			$x = 0.0015 \sin 2090t$	1 (for expression)
			or $x = 0.0015 \cos 2090t$	
		(ii)	any two max or -max	1
		(iii)	moving upwards	1
	(c)	(i)	$I_0 = V_0/R = 1.5 / 6.0 = 0.25 \text{ mA}$	1
			$I = 0.0025 \sin 2090t$	1
			or $I = 0.0025 \cos 2090t$	
		(ii)	$B = \mu_0 I / 2\pi d$	
			$= 2.5 \times 10^{-5} \text{ T}$	1
		(iii)	induced current flows from B to A and induced B-field direction according to Right-hand grip rule is anticlockwise as viewed from point A or clockwise as viewed from point B.	1
	(d)		The magnetic field of the induced current generates a magnetic force which acts to slow the motion of the wire.	1